1 Abstract

Classical *Schrödinger Bridge* (SB) problem seeks a path measure that minimally deviates from a given reference measure subject to prescribed initial and terminal marginals. This works investigate numerical approach to dynamic of interacting brownian particle with normal reflection term in low-dimensional setting ($d \leqslant 3$) on bounded domain $\mathcal{D} \subset \mathbb{R}^d$. To alleviate the difficulty due to nonlinearity incurred by non-local coupling, a nested fixed-point is adopted with Sinkhorn-type iterates. Further, fully semi-lagrangian discretization is used to propagate the solution of forward-backward partial differential equations (FB-PDEs).

2 Introduction

3 Related Works

Interest in spatially-constrained SB is largely under-developed, especially when also incorporating mean-field framework. This work is inspired by [4, 11]. First paper studies SB with reflected dynamic with Sinkhorn iterate where the time marching solution involve Neuman and Rudin boundary conditions, and show great performance in low-dimension. The latter generalized standard Sinkhorn to accept partial-integro-PDEs associated with mean-field SB resulting in nested version. Along this line [1] studies degenerate, sub-Riemannian dynamics, and use Sinkhorn simiarl to the former. Recent work [16] use Finite element discretization to solve the weak form of coupled PDEs for reflected SB. [9] derives forward-backward SDEs associated coupled FB-PDEs of reflected SB and extend divergence-based likelihood training in smooth bounded domain and shown to scale in high-dimension. Discrete-time framework also been explored with chance-constrants [18, 19] with mixture of Guassian marginal contraints. Alternative method such as introducing direct soft-penalty for particle escaping region exists as demonstrated in [15] using FB-SDE representation. There is a direct link between system of optimality of SB and MFG, therefore one can naturally adopt a scheme in MFG such as finite-difference to solve SB problem. [5–7] adopt a semi-lagrangian approach.

4 Problem Formulation

Let $\mathcal{D} \subset \mathbb{R}^d$ and boundary $\partial\mathcal{D}$, represented by smooth level-set $\chi(t,x)$ via $\chi < 0$ on $\mathcal{D}$ and $\chi = 0$ on $\partial\mathcal{D}$. Assuming $\nabla\chi \neq 0$ on $\partial\mathcal{D}$, the outward unit normal $\mathbf{n}(x)$ is obtained by $\mathbf{n} = \nabla\chi/\|\nabla\chi\|$. Denote by $\mathscr{P}(\mathcal{S})$ the space of probability measures on $\mathcal{S}$ endowed with the 2-Wasserstein metric $W_2(\cdot,\cdot)$ and having finite 2nd-moment. Let $\Omega :- \mathcal{C}(\mathcal{T};\mathbb{R}^d)$ where $\mathcal{T} :- [0,T]$ be canonical path space. Let $\mathbb{P}_t \in \mathscr{P}(\mathbb{R}^d)$ be t-time pushforward marginal of $\mathbb{P} \in \mathscr{P}(\Omega)$. For given fixed $\mathbb{P}$ and $\mu_{\text{init}} \in \mathscr{P}(\mathbb{R}^d)$, consider the dynamic of weakly-interacting brownian particle modeled by Mckean-Vlaskov reflected SDEs:

$$dX_t = -(\nabla E \star \mathbb{P}_t)(X_t) + \sigma\ dW_t - \mathbf{n}_t dV_t \qquad , X_0 \sim \mu_{\text{init}} \tag{1}$$

which induces the path law $\Gamma(\mathbb{P}) :- \text{Law}\left((X_t)_{t\in\mathcal{T}}\right)$, W is $\mathbb{R}^d$-valued Wiener on $(\Omega, \mathcal{F}, \mathbb{F} = (\mathcal{F}_t)_{t\in\mathcal{T}}, \mathbb{P}_0)$ Here E is potential, $(\nabla E \star \mu)(\cdot) :- \int_{\mathbb{R}^d} \nabla E(\cdot - y)\ \mu(dy)$. And V is a continuous, non-decreasing, $\mathbb{F}$-adapted bounded-variation process satisfying $V_0 = 0$, $\int_0^T \mathbf{1}_{\{X_s \notin \partial\mathcal{D}_s\}}\ dV_s = 0$, and $V_t = \int_0^t \mathbf{1}_{\{X_s \in \partial\mathcal{D}_s\}}\ dV_s$. For simplicity takes diffusion be $\sigma \equiv \sqrt{2\kappa}\mathbb{I}_d$ for $\kappa_0 > 0$. In [] shows that the rate function governing the large deviations for mean-field type dynamic in (1) is $H(\mathbb{P} \parallel \Gamma(\mathbb{P}))$. Unlike standard non-mean-field SB in that the referenc measure is no longer fixed and the convexity property is losed due to nonlinear dependency in its arguments of relative entropy []. Under appropriate hypotheses, the existence of $\mathbb{F}$-progressively measurable $\alpha(t,x)$ is guaranteed by Girsanov's, the relative entropy can be parametrized by process α:

$$H\big(\mathbb{P} \parallel \Gamma(\mathbb{P})\big) = \frac{1}{2}\mathbb{E}^{\mathbb{P}}\left[\int_0^T \|\alpha(t,X_s)\|^2\ ds\right] \tag{2}$$

and that X_t follows the controlled Mckean Vlasov SDE:

$$dX_t = b^\alpha(t, X_t, \mathbb{P}_t)\ dt + \sigma_t\ dW_t^{\mathbb{P}} - \mathbf{n}_t dV_t, \quad X_0 \sim \mu_{\text{init}}, \tag{3}$$

where $W^{\mathbb{P}}$ is a standard $(\mathbb{P}, \mathcal{F}_t)$-Brownian, $b^\alpha(t,x,\mu) :- \sigma_t\alpha_t - \nabla E \star \mu$. Further, under additional hypothesis, the perturbation is markovian feedback control. i.e. there exists $\Phi_t \in H_{-1}(\rho_t : t \in \mathcal{T})$ such that $\alpha_t = \Phi_t(X_t)$, $dt \times d\mathbb{P}$ a.s. (here assuming the control dimension is similar to state dimension). Following [], using transformed control $m := \rho\alpha \in \mathbb{R}^d$, rewrite (2) as SOC problem

$$\mathcal{J}^*_{\text{rmf-sb}} = \inf_{(\alpha,\rho)} \mathcal{J}_{\text{rmf-sb}}(m,\rho), \text{ where } \mathcal{J}_{\text{rmf-sb}}(m,\rho) :- \frac{1}{2}\int_0^T \int_{\mathbb{R}^d} \|m_t\|^2/\rho_t dx\ dt. \tag{4}$$

The law of sample path $(X_t)_{t\in\mathcal{T}}$ in (3) can be characterized as solving the martingale problem, i.e. under suitable assumption, for any $\psi \in \mathcal{C}^2(\mathcal{D})$, the process

$$M_t := \psi(X_t) - \psi(X_0) - \int_0^t \mathcal{G}[\rho_s]\psi(X_s)ds \tag{5}$$

is $\mathbb{P}$-local martingale with respect to $\mathbb{F}$ where $\mathcal{G}$ is generator for (3) defined by $\mathcal{G}[\rho] := \kappa\nabla\rho + \nabla\rho\cdot(\sigma\alpha - E\star\rho)$. It follows that [], the marginal flows $(\rho_t)_{t\in\mathcal{T}}$ is a weak solution of forward Kolmogorov equation:

$$\begin{aligned} \partial_t\rho + \nabla\cdot J = 0,\ (t,x)\in\mathcal{D},\\ J\cdot\mathbf{n} = 0,\ (t,x)\in\partial\mathcal{D} \end{aligned} \tag{6}$$

where probability flux is $J(m,\alpha) = \sqrt{2\kappa}m - \rho\nabla E\star\rho - \kappa\nabla\rho$, where here using $m := \rho\alpha$. One can now characterize the 1st-order optimality condition associated with (4) with contraint (6) which yields a system of forward-backward PDEs. Here the system is adjacent to that appear in [], supplemented with additional Neuman and Robin boundary conditions. The proof given here is straightforward application of Lagrange mulitplier on Banach space.

4.1 Assumption

- The domain $\mathcal{D}\subset\mathbb{R}^d$ is sufficiently regular.
- Interacting term $E\in\mathcal{C}^2\left(\mathbb{R}^d;\mathbb{R}\right)$ is symmetric, i.e. $E(\cdot) = E(-\cdot)$ with $\sup_{z,v\in\mathbb{R}^d} v^\top\nabla^2E(z)v < +\infty$.
- The admissible controls $\alpha\in\Theta_\alpha := H^1(\rho)$ s.t. $\exists\psi\in H^1$ with $\alpha = \nabla\psi$, $|m|^2/\rho\in L^1$, and $\rho\in\mathcal{C}\left(\mathcal{T},\mathscr{P}(\mathcal{D})\right)\cap L^2\left(0,T;H^1(\mathcal{D})\right)$ with $\rho\geqslant 0$, $\int_{\mathcal{D}}\rho = 1$ and $\rho(0,\cdot)dx = \mu_{\text{init}}$, $\rho(T,\cdot)dx = \mu_{\text{targ}}$

The rest of this work focus solving the following coupled FB-PDEs which can be drived in straightforward manner by mean of Lagrange Multiplier on Banach space. The difference between (PDE-opt) and that which given by [2] is the extra boundary condition.

Proposition 1. *The optimal control α^*,ρ^* are given by $\alpha^* = \sqrt{2\kappa}\nabla\log\varphi$ where $(\phi,\rho)\to(\varphi,\hat\varphi) := (e^{-\phi},\rho\exp(E\star\rho/\kappa+\phi))$ solve the following FB-PDEs:*

$$\begin{cases} -\partial_t\varphi = \left(\mathcal{A}^\rho + \mathcal{R}^\rho\right)[\varphi] \\ \partial_t\hat\varphi = \left(\mathcal{A}^{\rho,\dagger} + \mathcal{R}^{\rho,\dagger}\right)[\hat\varphi] \end{cases}, \text{ on } (0,T)\times\mathcal{D}, \quad \text{such that } \begin{cases} \nabla\varphi^\top\mathbf{n} = \nabla\hat\varphi^\top\mathbf{n} = 0,\ \text{on } (0,T)\times\partial\mathcal{D} \\ \mu_{init} = W[\rho]\varphi_0\hat\varphi_0,\ \text{on } \{0\}\times\mathcal{D} \\ \mu_{targ} = W[\rho]\varphi_T\hat\varphi_T,\ \text{on } \{T\}\times\mathcal{D} \end{cases} \tag{7}$$

where $\mathcal{A}^\rho[\eta] := (\nabla E\star\rho)\cdot\nabla\eta - \kappa\Delta\eta$ is advection-diffusion operator and $\mathcal{R}^\rho[\eta] := -\eta\nabla E\star(\rho\nabla\log\eta)$ is nonlocal source term, and $W[\rho](t,x) := e^{-E\star\rho/\kappa}$.

4.2 Fixed-Point Iterates

The constructive scheme for the PDE-opt solver is obtained via a single-loop Picard-Sinkhorn iteration to find the fixed point of the flow $\rho = (\rho_t)_{t\in\mathcal{T}}\in\mathcal{C}(\mathcal{T}\mathscr{P}(\mathcal{D}))$. We define the marginal map F and the inner propagator G by

$$F[\rho;\varphi,\hat\varphi] := W[\rho]\varphi\hat\varphi \quad\text{and}\quad G[\varphi_T;\mu,\varphi,\hat\varphi] := \left(M_T^\mu\mathcal{S}_{0\to T}^{\dagger,\mu,\hat\varphi}\circ M_0^\mu\mathcal{S}_{T\to 0}^{\mu,\varphi}\right)[\varphi_T]. \tag{8}$$

The propagator G shares a very similar structure with the classical linear Schrödinger bridge (SB), which is a well-studied numerical method [8, 14]. Here, $\mathcal{S}_{0\to T}^{\dagger,\mu,\hat\varphi}$ and $\mathcal{S}_{T\to 0}^{\mu,\varphi}$ are respectively the backward and forward time-split propagators of the linearized PDE-opt system frozen at $(\mu,\varphi,\hat\varphi)$. The term "linearized PDEs" refers to the fact that under fixed $(\rho^{n-1},\varphi^{n-1},\hat\varphi^{n-1})$, both the nonlocal drift $b_t^{n-1} := -\nabla E * \rho_t^{n-1}$ and the reaction rate $r_t^{n-1} := -\nabla E * (\rho_t^{n-1}\nabla\log\varphi_t^{n-1})\in L^\infty(\mathcal{D})$ are frozen in time and space, reducing the nonlinear forward-backward PDEs to standard non-autonomous linear Cauchy problems with the time-varying generator $\mathcal{A}_t^{n-1} + \mathcal{R}_t^{n-1}$. It follows that this generator induces a unique backward evolution family $(\mathcal{P}_{t\to s}^{n-1})_{0\leqslant s\leqslant t\leqslant T}$ on $L^2(\mathcal{D})$; see also [3]. Here, M_T^μ and M_0^μ are Sinkhorn scaling operators defined pointwise by

$$\hat\varphi_0(\cdot)\xleftarrow{M_0^\mu}\frac{\mu_{\text{init}}(\cdot)}{W[\mu](\cdot,0)\varphi(\cdot,0)} \quad\text{and}\quad \hat\varphi_T(\cdot)\xleftarrow{M_T^\mu}\frac{\mu_{\text{targ}}(\cdot)}{W[\mu](\cdot,T)\varphi(\cdot,T)}. \tag{9}$$

Note that these two operators are similar to those in [11]. We suppress the superscripts ρ^{n-1},φ^{n-1} for readability. For a uniform temporal subdivision $0 = t_0 < t_1 < \cdots < t_{N_T} = T$ of step size $\Delta t = t_{k+1} - t_k$, we define the exact continuous symmetric Strang splitting $\mathcal{S}_{t_{k+1}\to t_k}$ acting on φ via the composition

$$\varphi(t_k,\cdot)\approx\mathcal{S}_{t_{k+1},t_k}[\varphi](t_{k+1},\cdot) := \left(\mathcal{R}_{t_{k+\Delta t/2}\to t_k}\circ\mathcal{U}_{t_{k+1}\to t_k}\circ\mathcal{R}_{t_{k+1}\to t_{k+\Delta t/2}}\right)[\varphi](t_{k+1},\cdot). \tag{10}$$

Since $\mathcal{R}[\eta](s,x) = r_s(x)\eta(x)$ is a bounded pointwise multiplication operator on $L^2(\mathcal{D})$, it admits an analytical solution of the form $\mathcal{R}_{t\to s}[\eta](x) = \eta(x)\exp\int_s^t r_u(x)\,du$, whose half-step approximations via quadrature points are given by

$$\tilde{\mathcal{R}}_{k+1\to k+\Delta t/2}[\eta] :- \eta e^{\Delta t r_{t_{k+1}}/2} \quad \text{and} \quad \tilde{\mathcal{R}}_{k+\Delta t/2\to k}[\eta] :- \eta e^{\Delta t r_{t_k}/2}. \tag{11}$$

Next, the middle operator $\mathcal{U}_{t_{k+1}\to t_k}$ solving the advection-diffusion subproblem has the Feynman-Kac representation [17] over $[t_k, t_{k+1}]$ with Neumann boundary conditions given by

$$\mathcal{U}_{t_{k+1},t_k}[\eta](x) :- \mathbb{E}^{\mathbb{P}}\left[\eta(t_k, X_{t_k}) \mid X_{t_{k+1}} = x\right] = \lim_{K\to\infty} \sum_{1\leqslant m\leqslant K} w_m \eta\left(t, X^x_{t_k,t_{k+1}}(\omega_m)\right), \tag{12}$$

where $X_{t_k,t_{k+1}}$ solves (for fixed ρ) the forward process starting from $X_{t_k} = x$, and w_m are normalized positive weights. Due to random noise incurred by finite approximations of expectations via Monte Carlo (of order $O(\sqrt{\Delta t/K})$), a deterministic cubature stencil is used instead, known as the Semi-Lagrangian (SL) method. Defining $\mathcal{S}_{T\to 0} :- \mathcal{S}_{t_1\to t_0}\circ\mathcal{S}_{t_2\to t_1}\circ\cdots\circ\mathcal{S}_{t_{N_T}\to t_{N_T-1}}$, it is guaranteed [3] that a second-order error bound for sufficiently smooth terminal data φ_T holds; i.e., there exists $C \geqslant 0$ independent of Δt such that

$$\|\mathcal{P}_{0,T}\varphi_T - \mathcal{S}_{T\to 0}\varphi_T\|_{L^2} \leqslant C\Delta t^2 \|\varphi_T\|_{L^2}. \tag{13}$$

Similarly, since the Fokker-Planck equation for $\hat{\varphi}$ is the formal L^2-adjoint of the backward equation, $\mathcal{S}^\dagger_{k\to k+1} :- (\mathcal{S}_{k+1\to k})^\dagger$ is obtained simply by reversing the operation:

$$\hat{\varphi}(t_{k+1},\cdot) \approx \mathcal{S}^\dagger_{t_k\to t_{k+1}}[\hat{\varphi}](t_k,\cdot) = \left(\mathcal{R}^\dagger_{t_{k+1}\to t_{k+\Delta t/2}}\circ\mathcal{U}^\dagger_{t_{k+1}\to t_k}\circ\mathcal{R}^\dagger_{t_{k+\Delta t/2}\to t_k}\right)[\hat{\varphi}](t_k,\cdot). \tag{14}$$

Notice that $\mathcal{R}^\dagger_{t\to s} = \mathcal{R}_{t\to s}$ is self-adjoint. Upon spatial discretization, this structure ensures the discretized forward propagator $\mathcal{S}_{0\to T}$ can be obtained via mass-weighted matrix transpose of $\mathcal{S}_{T\to 0}$; see also [12].

Algorithm 1: Fixed-Point Iteration Solver

Input: $\mu_{\text{init}}, \mu_{\text{targ}} \in \mathbb{R}^{N_x}_{>0}$, $\Delta t = T/N_T$, $\theta \in (0,1]$, threshold $\epsilon > 0$.
Initialize: $\boldsymbol{\varphi}^0_k, \hat{\boldsymbol{\varphi}}^0_k \leftarrow \mathbb{1}_{N_x}$ and $\boldsymbol{\rho}^0 \leftarrow \mathbb{1}_{N_x}/\operatorname{Vol}_d(\mathcal{D})$ for $k = 0,\ldots,N_T$. **for** $n = 1,\ldots,N_{iter}$ **do**
  Compute $\mathbf{V}^{n-1}_k, \mathbf{W}^{n-1}_k, \mathbf{R}^{n-1}_k, \mathbf{r}^{n-1}_k, \mathbf{b}^{n-1}_{k,l}$ for $k = 0,\ldots,N_T$
  Set $\boldsymbol{\varphi}^n_{N_T} \leftarrow \boldsymbol{\varphi}^{n-1}_{N_T}$
  for $k = N_T-1, N_T-2,\ldots,0$ **do**
    $\boldsymbol{\varphi}^n_k \leftarrow \mathbf{R}^{n-1}_k \boldsymbol{\Pi}^{n-1}_k \mathbf{R}^{n-1}_{k+1}\boldsymbol{\varphi}^n_{k+1}$
  Set $\hat{\boldsymbol{\varphi}}^n_0 \leftarrow \boldsymbol{\mu}_{\text{init}} \oslash \left(\operatorname{diag}(\mathbf{W}^{n-1}_0)\odot\boldsymbol{\varphi}^n_0\right)$
  for $k = 0,\ldots,N_T-1$ **do**
    $\hat{\boldsymbol{\varphi}}^n_{k+1} \leftarrow \mathbf{R}^{n-1}_{k+1}\boldsymbol{\Pi}^{n-1,\dagger}_k \mathbf{R}^{n-1}_k \hat{\boldsymbol{\varphi}}^n_k$
  Set $\boldsymbol{\varphi}^n_{N_T} \leftarrow \boldsymbol{\mu}_{\text{targ}} \oslash \left(\operatorname{diag}(\mathbf{W}^{n-1}_{N_T})\odot\hat{\boldsymbol{\varphi}}^n_{N_T}\right)$
  Compute $\boldsymbol{\rho}^{1/2}_k = \operatorname{diag}(\mathbf{W}^{n-1}_k)\odot\boldsymbol{\varphi}^n_k\odot\hat{\boldsymbol{\varphi}}^n_k$ and normalize in $L^1(\mathcal{D}_\Delta)$: $\boldsymbol{\rho}^{1/2}_k \leftarrow \boldsymbol{\rho}^{1/2}_k/(\mathbf{1}^\top M_{\text{lump}}\boldsymbol{\rho}^{1/2}_k)$
  Set $\boldsymbol{\rho}^n_k = (1-\theta)\boldsymbol{\rho}^{n-1}_k + \theta\boldsymbol{\rho}^{1/2}_k$ for $k = 0,\ldots,N_T$
  if $\max_{1\leqslant k\leqslant N_T-1}\|\boldsymbol{\rho}^n_k - \boldsymbol{\rho}^{n-1}_k\|_\infty \leqslant \epsilon$ **then**
    break

Unlike the nested two-loop scheme in [10, 12, 13], Solver 1 executes only a single loop, and the nonlocal drift and reaction terms are updated alongside the scaling operators M_0 and M_T. This structure closely mirrors fictitious play and Picard iteration in mean-field games (MFG) [**?**] with extended Sinkhorn scaling applied to PDEs. The next section concerns spatial discretization via a monotone P_1 finite element Semi-Lagrangian scheme.

4.3 Semi-Lagrangian Discretization

What remains now is spatial discretization via SL. As mentioned in [5], the SL scheme is monotone, unconditionally stable, and positivity-preserving without requiring the parabolic CFL condition $\Delta t = O(\Delta x^2)$. Let $\mathcal{D}_\Delta = \{x_i\}_{1\leqslant i\leqslant N_x} \subset \overline{\mathcal{D}}$ be the vertices of a simplicial triangulation of size Δx over the polyhedral domain $\mathcal{D}$, and let $\mathcal{T}_\Delta = \{t_k = k\Delta t\}_{0\leqslant k\leqslant N_t}$ be the time subdivision. At t_k, the continuous field $\eta \in L^2(\mathcal{D})$ is approximated by nodal values $\boldsymbol{\eta}_k = (\eta_{k,1},\ldots,\eta_{k,N_x}) \in \mathbb{R}^{N_x}$. Its continuous reconstruction is obtained via the P_1 finite element interpolator $\mathcal{I} : \mathbb{R}^{N_x} \to \mathcal{C}_b(\mathcal{D})$ defined by $\mathcal{I}[\eta](\cdot) :- \boldsymbol{\beta}(\cdot)^\top\boldsymbol{\eta}$, where $\boldsymbol{\beta} = (\beta_1,\ldots,\beta_{N_x})^\top$ is the vector of standard piecewise linear Lagrange basis functions satisfying $\beta_i(x_j) = \delta_{ij}$ and $\mathbf{1}^\top\boldsymbol{\beta} = 1$. This

Algorithm 2: Solver 1: Fixed-Point Iteration

for $n = 1, \dots, N_{iter}$ **do**

- $\varphi^n \leftarrow \mathcal{S}_{T\to 0}\left[\varphi_T^{n-1}; \rho^{n-1}, \varphi^{n-1}\right]$ `// backward propagate`
- $\hat{\varphi}_0^n \leftarrow M_0^{\rho^{n-1}}[\varphi^n]$ `// initial marginal scale`
- $\hat{\varphi}^n \leftarrow \mathcal{S}_{0\to T}^{\dagger}\left[\hat{\varphi}_0^n; \rho^n, \hat{\varphi}^n\right]$ `// forward propagate`
- $\varphi_T^n \leftarrow M_T^{\rho^{n-1}}[\hat{\varphi}^n]$ `// terminal marginal scale`
- $\rho^n \leftarrow (1-\theta)\rho^{n-1} + \theta F\left[\rho^{n-1}; \varphi^n, \hat{\varphi}^n\right]$ `// density update`

interpolation introduces a spatial discretization error strictly bounded by $O(\Delta x^2)$. Following [7], we choose $\beta_i(x) = \max\{0, 1 - |x - x_i|/\Delta x\}$, which has support on $[x_i - \Delta x, x_i + \Delta x]$. The choice of linear interpolants (yielding $O(\Delta x^2)$ error) over higher-order alternatives such as Hermite polynomials or splines (which achieve spatial errors up to $O(\Delta x^4)$) is motivated by properties endowed by $\mathcal{I}$, such as positivity preservation, monotonicity, and computational efficiency.

We approximate $\mathcal{U}_{t_{k+1}\to t_k}$ with a second-order weak $(2d+1)$-point stencil; see [5]. For node $x_i \in \mathcal{D}_\Delta$ at time t_k, the advective transport under reference drift $\mathbf{b}_{k,i}^{n-1} := (-\nabla E \star \rho_k^{n-1})(x_i) \in \mathbb{R}^d$ with diffusivity $2\kappa\mathbb{I}_d$ is approximated by the unconstrained backward characteristic path over $[t_k, t_{k+1}]$ via

$$y_{k,i,l}^s(a) = x_i - \Delta t \mathbf{b}_{k,i}^{n-1} + e_l s\sqrt{6\kappa d\Delta t}, \quad s \in \{0, \pm 1\},\ l \in \{1, \dots, d\}, \tag{15}$$

where e_l is the canonical basis vector. Neumann reflection boundary conditions are enforced for any characteristic exiting $\mathcal{D}$, following [5] for the special case of normal reflection. Specifically, recalling that $\chi < 0 \iff x \notin \mathcal{D}$ denotes the signed distance function, the Skorokhod map reads

$$\hat{y}_{k,\mathbf{i},\ell}^s := \begin{cases} y_{k,\mathbf{i},\ell}^s & \text{if } y_{k,\mathbf{i},\ell}^s \in \mathcal{D}, \\ y_{k,\mathbf{i},\ell}^s - 2\chi(y_{k,i,l}^s)\mathbf{n}(\chi(y_{k,i,l}^s)) & \text{if } y_{k,\mathbf{i},\ell}^s \notin \mathcal{D}, \end{cases} \tag{16}$$

where $\mathbf{n}(\chi(a)) = \nabla\chi(a)/\|\nabla\chi(a\|_2$. Using these positions and three stencil weights, the fully discrete SL operator approximated at x_i is given by the transition matrix $\mathbf{\Pi}_k \in \mathbb{R}^{N_x\times N_x}$ via $\mathcal{U}_{t_{k+1}\to t_k}[\boldsymbol{\eta}](x_i) \approx \sum_{1\leqslant j\leqslant N_x}(\mathbf{\Pi}_k^{n-1})_{i,j}\eta_j$, where

$$(\mathbf{\Pi}_k)_{a,b} := \frac{2}{3}\beta_j(\hat{y}_{k,i,1}^0) + \frac{1}{6d}\sum_{\ell=1}^{d}\left(\beta_j(\hat{y}_{k,i,\ell}^+) + \beta_j(\hat{y}_{k,i,\ell}^-)\right). \tag{17}$$

This choice achieves second-order accuracy. For the increment $\Delta Y_{k,i} = Y_{k,i} - x_i$, we have $\mathbb{E}[\Delta Y_{k,i}] = -\Delta t\mathbf{b}_k^{n-1}(x_i)$ and $\mathrm{Cov}(\Delta Y_{k,i}) = 2\kappa\Delta t\mathbb{I}_d + O(\Delta t^2)$, which is congruent with $\mathbb{E}[dX_t]$ and $\mathrm{Cov}(\sqrt{2\kappa}dW_t)$ up to second order, respectively. Furthermore, by construction, the matrix $\mathbf{\Pi}$ is row-stochastic (i.e., non-negative with $\mathbf{\Pi}^\top\mathbf{1} = \mathbf{1}$). For a probabilistic interpretation of the SL scheme in terms of controlled Markov processes, see [5]. Finally, let $M_{\mathrm{lump}} = \mathrm{diag}(m_1, \dots, m_{N_x})$ with mass weights $m_i = \int_{\mathcal{D}}\beta_i(a)\,da > 0$, and let $\mathbf{D}_l \in \mathbb{R}^{N_x\times N_x}$ be the lumped-mass discretized partial derivative ∂_{x_l} along $l \in \{1, \dots, d\}$ defined by $(\mathbf{D}_l)_{i,j} = m_i^{-1}\int_{\mathcal{D}}\beta_i(x)\partial_{x_l}\beta_j(x)\,dx$.

The discretized objects are summarized below for iteration $n-1$ (dropping superscripts for clarity), step k, $l \in \{1, \dots, d\}$, and $i \in \{1, \dots, N_x\}$:

$$\begin{aligned} (\mathbf{V}_k)_i &:= \sum_j E(x_i - x_j)\rho_{k,j}m_j, & \mathbf{W}_k &:= \mathrm{diag}\left(\exp(-\mathbf{V}_k/\kappa)\right), \\ \mathbf{R}_k &:= \mathrm{diag}\left(e^{\Delta t\mathbf{r}_k/2}\right), & \mathbf{r}_k &:= -\sum_{l,j}\rho_{k,j}m_j\partial_{x_l}E(x_i - x_j)^\top\frac{(\mathbf{D}_l\boldsymbol{\varphi}_k)_j}{\varphi_{k,j}}, \\ (\mathbf{b}_{k,l})_i &:= -\sum_j \partial_{x_l}E(x_i - x_j)\rho_{k,j}m_j. \end{aligned} \tag{18}$$

With respect to the lumped $L^2(\mathcal{D}_\Delta)$ inner product $\langle\mathbf{u}, \mathbf{v}\rangle_{M_{\mathrm{lump}}} := \mathbf{u}^\top M_{\mathrm{lump}}\mathbf{v}$, the discrete adjoint of the single-step backward matrix operator $\mathbf{S}_{k+1\to k} := \mathbf{R}_k\mathbf{\Pi}_k\mathbf{R}_{k+1}$ is given explicitly by

$$(\mathbf{S}_{k+1\to k})^\dagger = M_{\mathrm{lump}}^{-1}(\mathbf{S}_{k+1\to k})^\top M_{\mathrm{lump}} = \mathbf{R}_{k+1}\mathbf{\Pi}_k^\dagger\mathbf{R}_k, \tag{19}$$

where $\mathbf{\Pi}_k^\dagger := M_{\mathrm{lump}}^{-1}\mathbf{\Pi}_k^\top M_{\mathrm{lump}}$ due to commutation with M_{lump}.

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